

Correlations and Lags Graphs

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely $I(1)$ -like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

corr_rows=296

lag_profiles_rows=872

Interpretation option: inspect strongest co-movements and dominant lag distances.

Table: Corr_Matrix

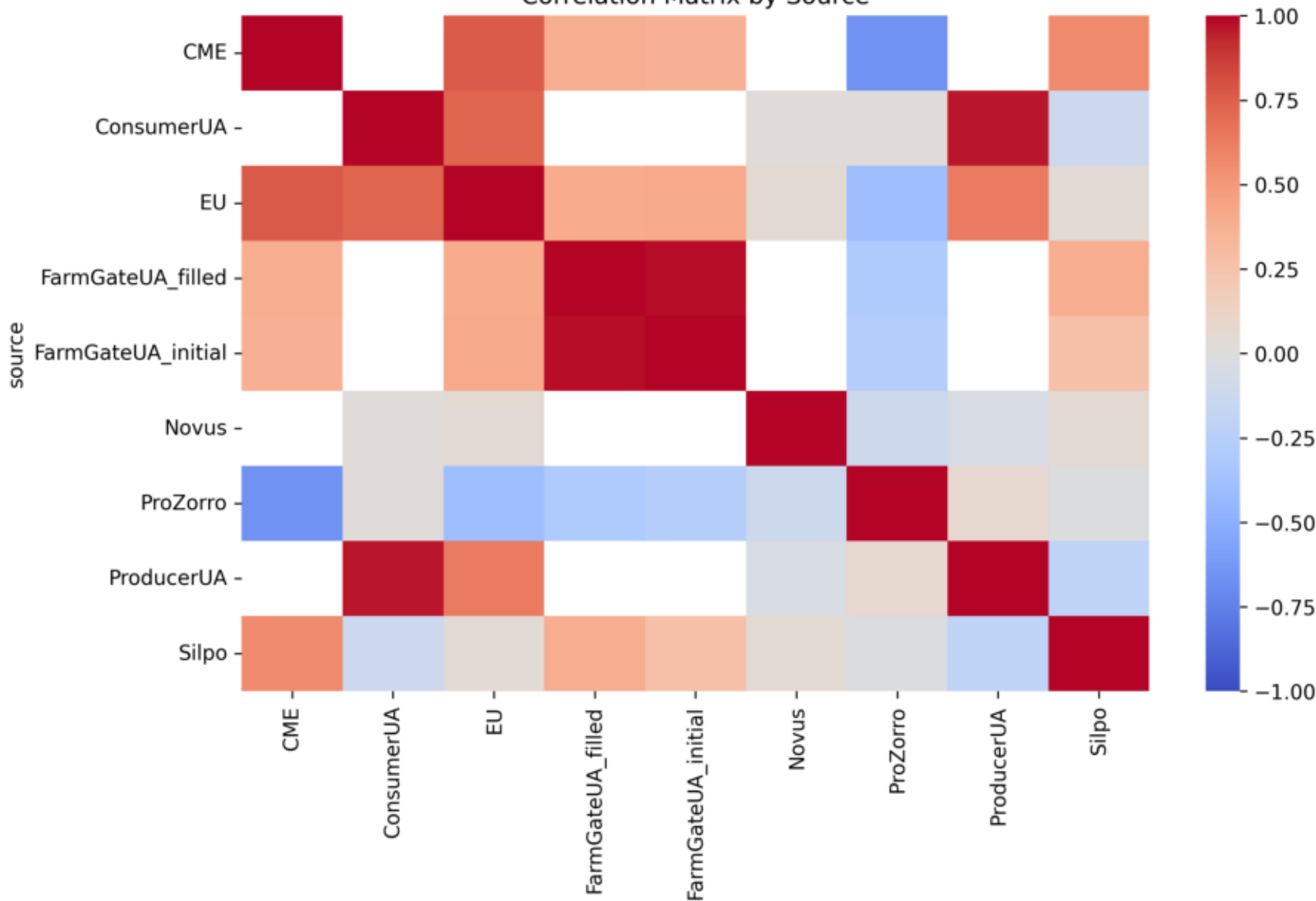
source	CME	ConsumerUA	EU	FarmGateUA_filled	FarmGateUA_initial	Novus	ProZorro	ProducerUA	Silpo
CME	1.0	nan	0.771150307014702	0.39636130313793916	0.3818437732242314	nan	-0.6686712585676096	nan	0.5739242968213558
ConsumerUA	nan	1.0	0.7331215385198419	nan	nan	0.02681338042040181	0.023872483772892125	0.9649157171256739	-0.12751976664861955
EU	0.771150307014702	0.7331215385198419	1.0	0.401960553922201	0.4146037200401722	0.05411886372768958	-0.39911500996322974	0.6367201119481288	0.04231622485451459
FarmGateUA_filled	0.39636130313793916	nan	0.401960553922201	1.0	0.9826279956426046	nan	-0.30096320420445527	nan	0.3955558075899114
FarmGateUA_initial	0.3818437732242314	nan	0.4146037200401722	0.9826279956426046	1.0	nan	-0.26589910003639705	nan	0.27660878164542574
Novus	nan	0.02681338042040181	0.05411886372768958	nan	nan	1.0	-0.12065550374558552	-0.043920496619733594	0.048276373219172226
ProZorro	-0.6686712585676096	0.023872483772892125	-0.39911500996322974	-0.30096320420445527	-0.26589910003639705	-0.12065550374558552	1.0	0.07811082527179705	-0.009897940313521869
ProducerUA	nan	0.9649157171256739	0.6367201119481288	nan	nan	-0.043920496619733594	0.07811082527179705	1.0	-0.21091575035180385
Silpo	0.5739242968213558	-0.12751976664861955	0.04231622485451459	0.3955558075899114	0.27660878164542574	0.048276373219172226	-0.009897940313521869	-0.21091575035180385	1.0

Table: Lag_Best

	product	standardized_type	pair_left	pair_right	lag_days	corr	is_best_lag
Сир	кисломолочний	cottage_cheese	Novus	Silpo	14	0.24523627303555948	1
	Вершки	cream	EU	ProZorro	28	-0.44201120383867126	1
	Вершки	cream	EU	ProducerUA	1	0.7270788393994388	1
	Вершки	cream	EU	Silpo	10	-0.6044676898703678	1
	Вершки	cream	Novus	Silpo	9	0.4556649937689249	1
	Вершки	cream	ProducerUA	Novus	18	-0.32517453059128315	1
	Вершки	cream	ProducerUA	ProZorro	20	-0.08491389732601401	1
	Вершки	cream	ProducerUA	Silpo	7	-0.4180798655236148	1
Сир	твердий	hard_cheese	EU	ProZorro	11	-0.26333701523095143	1
Сир	твердий	hard_cheese	EU	ProducerUA	27	0.686007360451993	1
Сир	твердий	hard_cheese	EU	Silpo	24	-0.4607038146699059	1
Сир	твердий	hard_cheese	Novus	Silpo	7	-0.4926293986262085	1
Сир	твердий	hard_cheese	ProducerUA	ConsumerUA	1	0.9861527112372024	1
Сир	твердий	hard_cheese	ProducerUA	Novus	30	-0.43688233284645367	1
Сир	твердий	hard_cheese	ProducerUA	ProZorro	30	-0.112574663105113	1
Сир	твердий	hard_cheese	ProducerUA	Silpo	16	-0.37374547140444553	1
	Кефір	milk	ProducerUA	Silpo	18	-0.36968876199061973	1
Молоко	питне	milk	EU	ProZorro	27	0.19798230710584866	1
Молоко	питне	milk	EU	ProducerUA	27	0.5301098856434086	1
Молоко	питне	milk	EU	Silpo	8	-0.5219797447000611	1
Молоко	питне	milk	Novus	Silpo	11	-0.4761644911108315	1
Молоко	питне	milk	ProducerUA	ConsumerUA	1	0.9068263834913155	1
Молоко	питне	milk	ProducerUA	Novus	14	0.5727037857042536	1
Молоко	питне	milk	ProducerUA	ProZorro	26	-0.07547451021848318	1
Молоко	питне	milk	ProducerUA	Silpo	15	-0.5247120883607513	1
Інше/невідомо	other		EU	ProZorro	21	-0.9242646162761372	1
Інше/невідомо	other		EU	Silpo	26	0.5175772654738634	1
	Сметана	sour_cream	ProducerUA	ConsumerUA	1	0.9940901205039061	1
	Сметана	sour_cream	ProducerUA	ProZorro	14	0.09401258045745899	1
	Сметана	sour_cream	ProducerUA	Silpo	14	-0.29764561987276605	1
	Йогурт	yogurt_dessert	Novus	Silpo	24	-0.4524200881519273	1

corr_matrix_sources.png

Correlation Matrix by Source



lag_best_bar.png

Best Lag (days) by Product/Pair

